GEOGRAPHY

PHYSICAL GEOGRAPHY

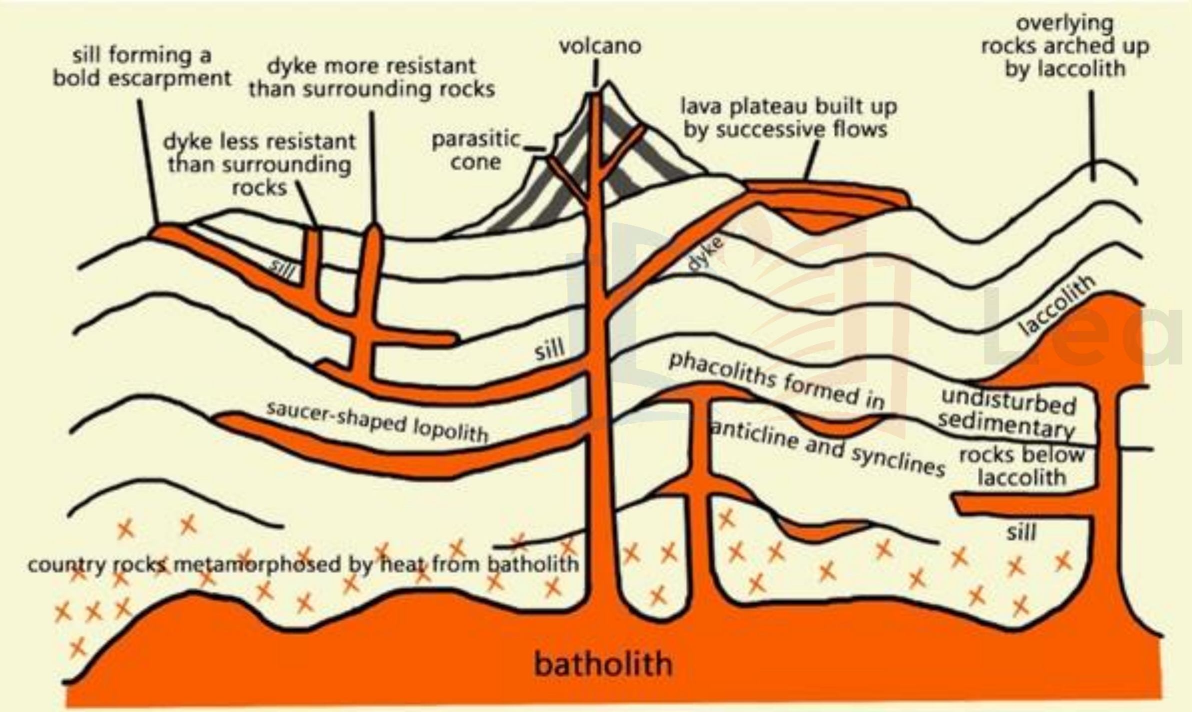
GEOMORPHOLOGY

MODULE - 21

VOLCANOES

- INTRUSIVE LANDFORMS

VULCANISM AND VARIOUS FORMS



HOW VARIOUS LANDFORMS ARE FORMED?

1. Magma while thrusting its way up to the surface may cool and solidify within the crust as plutonic rocks. It results in Intrusive Landforms.

2. Magmas that reach the surface and solidify form Extrusive Landforms.

VULCANISM AND VARIOUS FORMS

INTRUSION

It forms features such as dykes and sills. It can intrude into a low density rock such as sedimentary rock and when it cools into solid rock this intrusion is called a pluton. A pluton is an intrusion of magma that wells up from below the surface and they can form intrusive landforms.

EXTRUSION

The most familiar way for magma to escape is through lava. Lava eruptions can be fire fountains of liquid rock or thick slow moving rivers of molten material. Lava cools to form volcanic rock as well as volcanic glass.

INTRUSIVE LANDFORMS

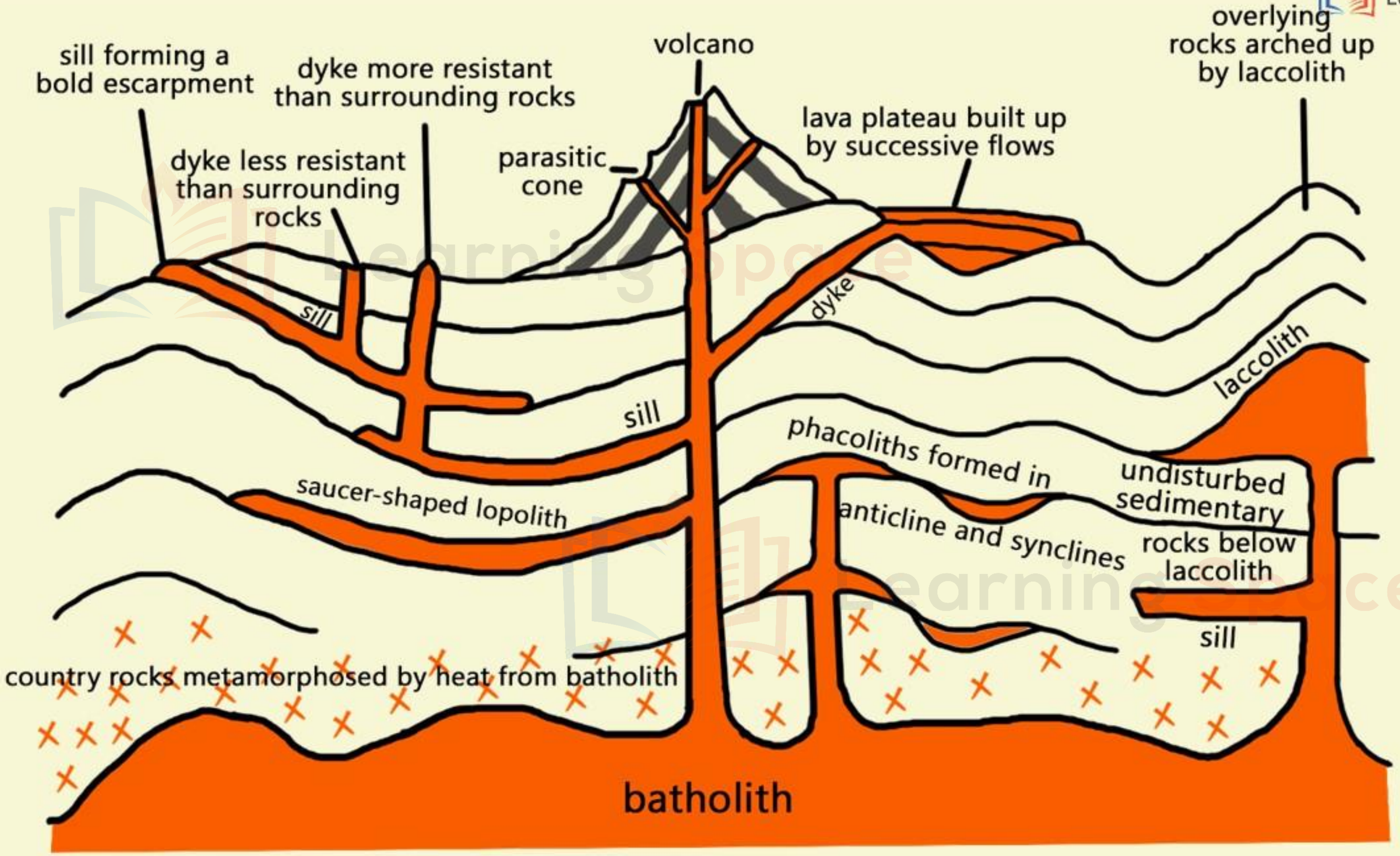
1 SILLS

1. The commonest intrusive landforms are Sills and Dykes.
2. When an intrusion takes place horizontally along the bedding plates of sedimentary rocks, it is known as Sill.
3. The denudation of the overlying sedimentary strata exposes the intrusion which will resemble a lava flow or form a bold escarpment.

INTRUSIVE LANDFORMS

2 DYKES

1. Similar intrusions when injected vertically as narrow walls of igneous rocks within the sedimentary layers are called Dykes.
2. Because of their narrowness, dykes seldom dominate the landscape.
3. When exposed to denudation, dykes may appear as upstanding walls or shallow trenches, depending on whether they are more or less resistant than the rocks in which they are placed.



sill forming a bold escarpment
dyke more resistant than surrounding rocks
volcano
overlying rocks arched up by laccolith
lava plateau built up by successive flows
parasitic cone
dyke less resistant than surrounding rocks
sill
dyke
laccolith
sill
phacoliths formed in anticline and synclines
undisturbed sedimentary rocks below laccolith
saucer-shaped lopolith
sill
country rocks metamorphosed by heat from batholith
batholith

SILLS

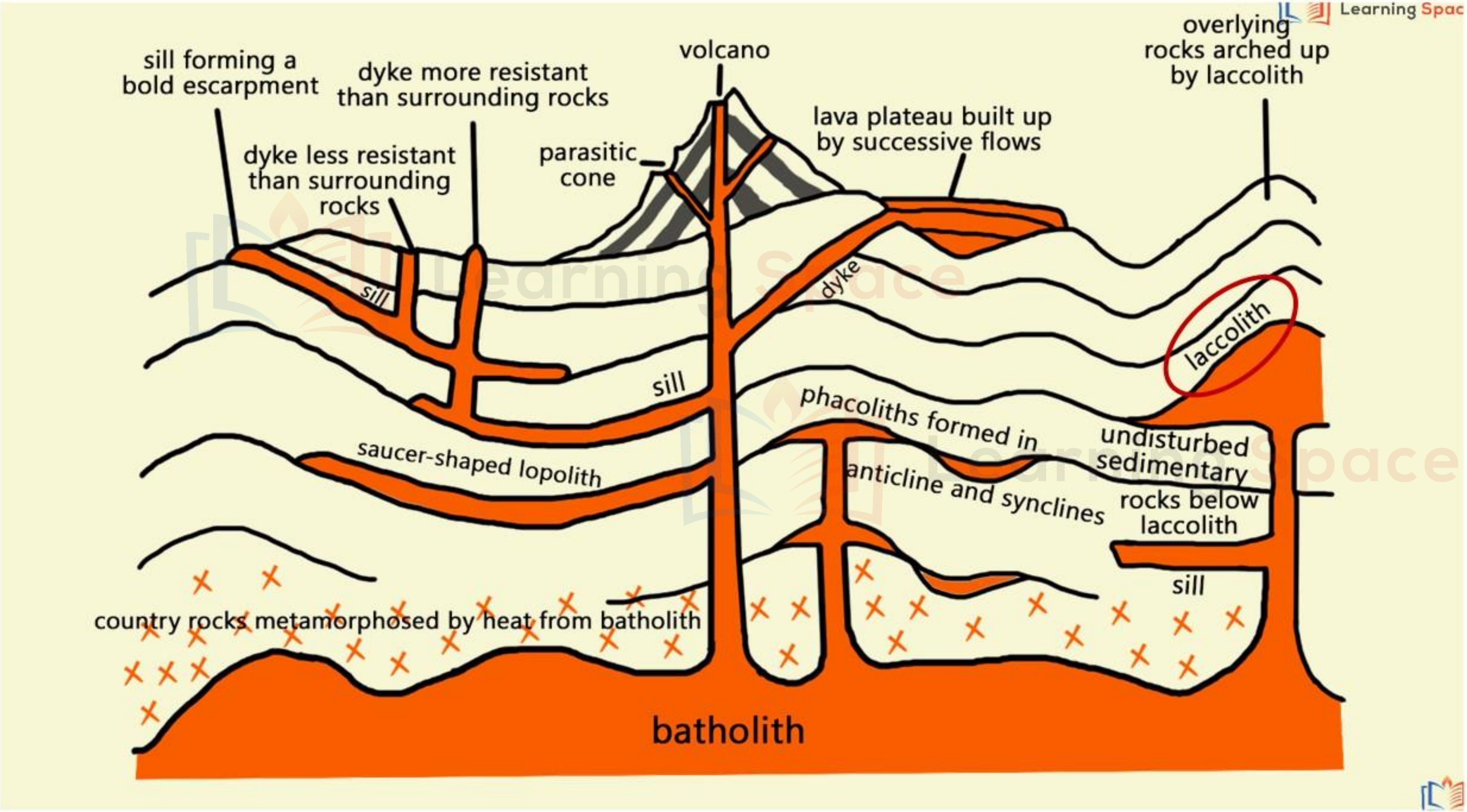
DYKES

INTRUSIVE LANDFORMS

3 LACCOLITHS

Henry Mountains in Utah

- It is a large blister or igneous mound with a dome-shaped upper surface and a level base fed by a pipe-like conduit from below.
- It arches up the overlying strata of sedimentary rocks.
- Examples are laccoliths of Henry Mountains, in Utah, United States of America.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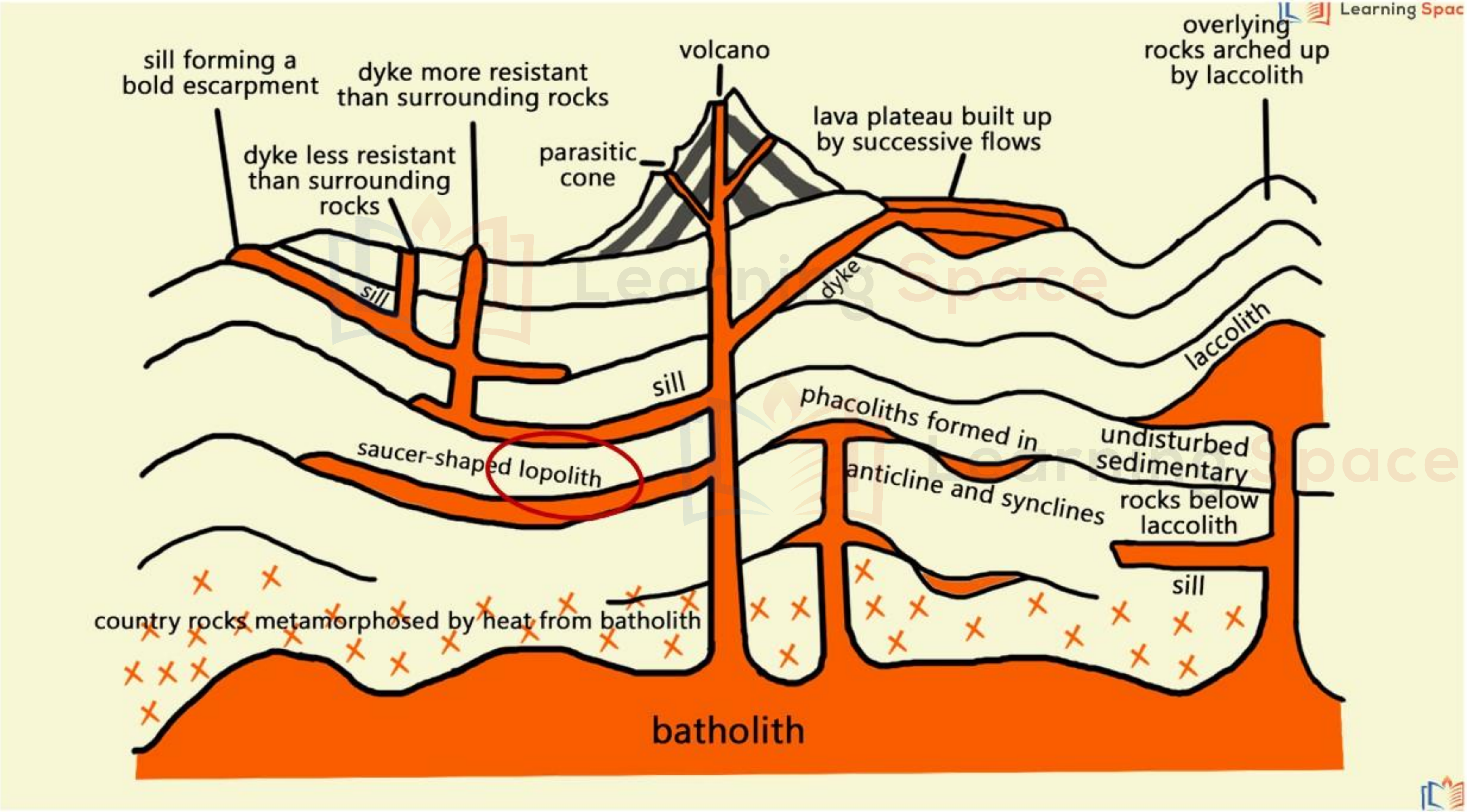
UNITED STATES
UTAH
HENRY MOUNTAINS
NORTH ATLANTIC OCEAN
NORTH PACIFIC OCEAN
GULF OF MEXICO
MEXICO
Learning Space
Learning Space



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INTRUSIVE LANDFORMS

4 LOPOLITH

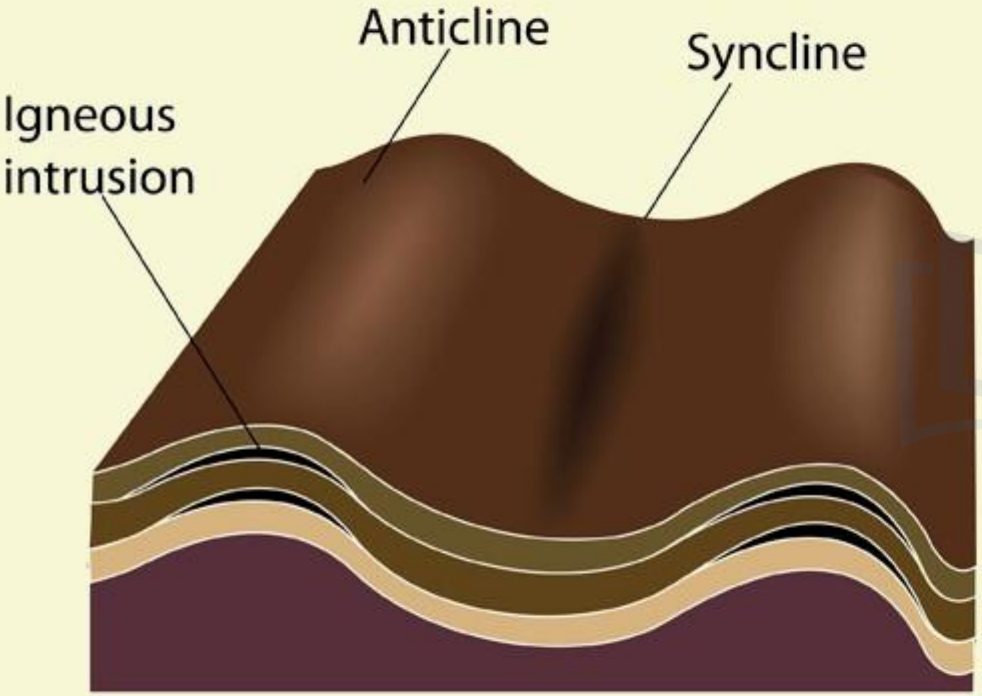
The Bushveld lopolith in southern Africa is several hundred kilometres across and contains the richest platinum deposits known.

- It is another variety of igneous intrusion with a saucer shape.
- A shallow basin is formed in the mid of the country rocks.
- The Bushveld lopoliths of Transvaal, South Africa are good examples.


BOTSWANA
BUSHVELD LOPOLITHS
NAMIBIA
TRANSVAAL
LESOTHO
SOUTH AFRICA
INDIAN OCEAN

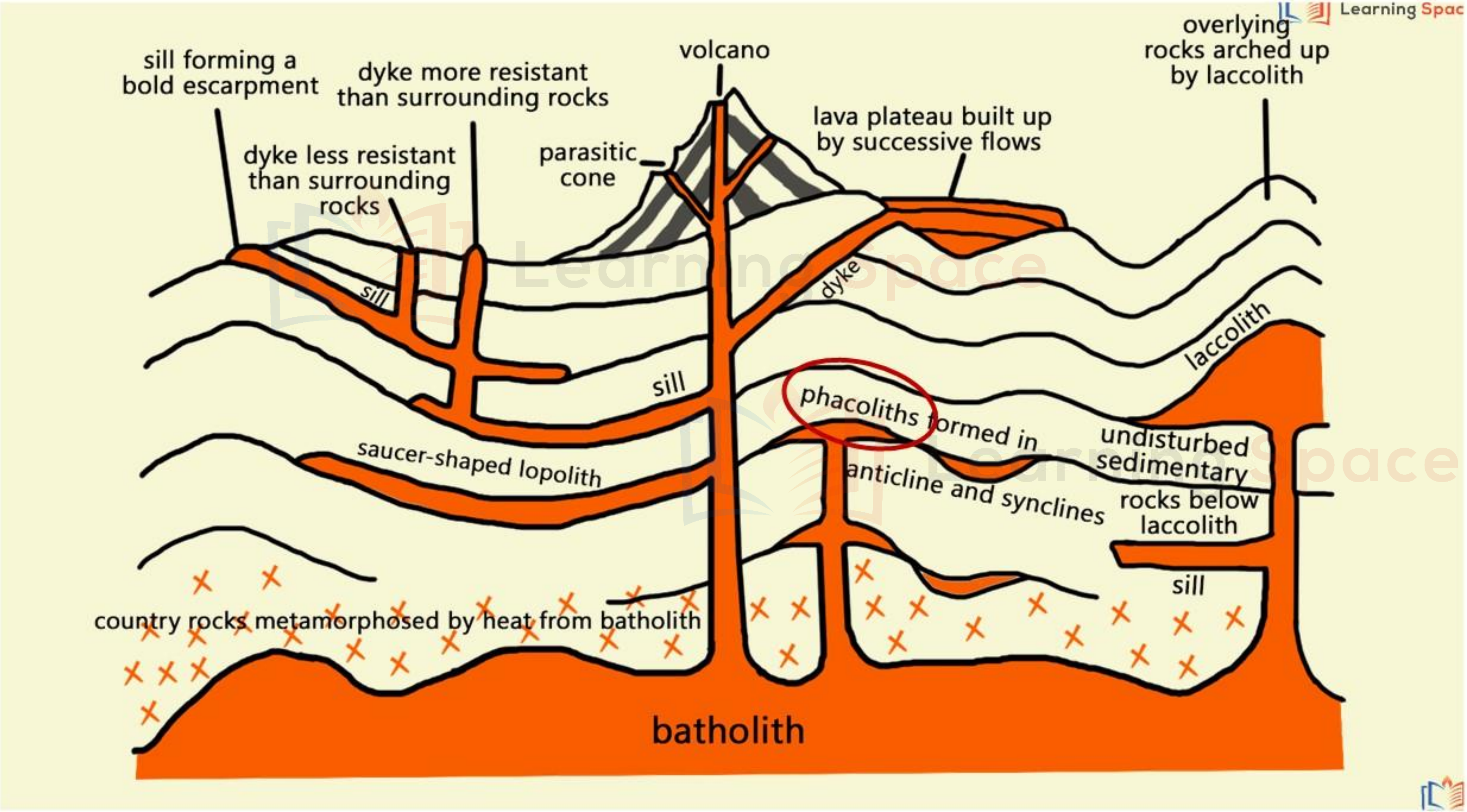
INTRUSIVE LANDFORMS

5 PHACOLITH



Phacoliths

- It is a lens-shaped mass of igneous rocks occupying the crest of an anticline or the bottom of a syncline and being fed by a conduit from beneath.
- Example is Corndon Hill in Shropshire, England.



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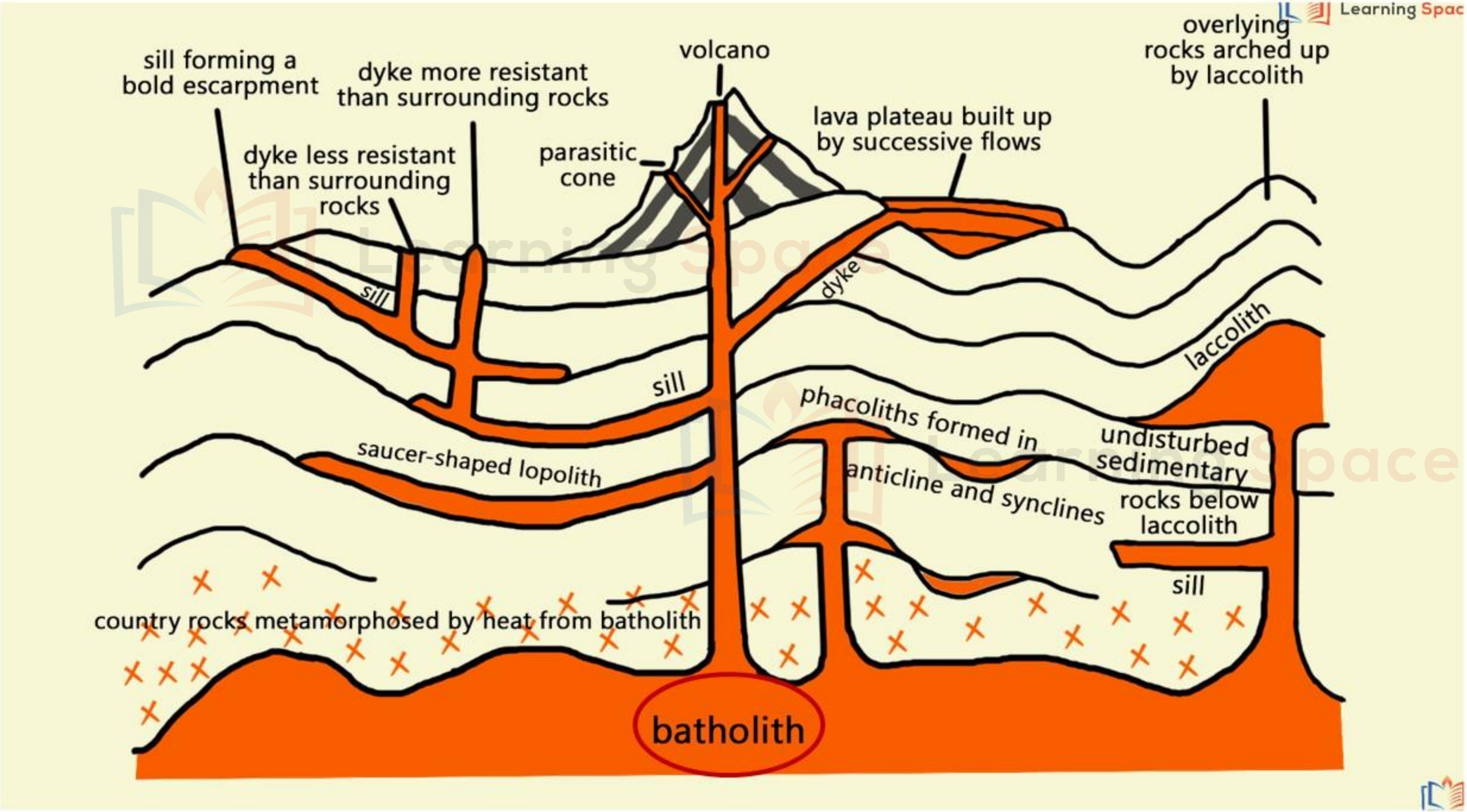
North Atlantic Ocean
North Sea
SCOTLAND
NORTHERN IRELAND
IRELAND
Corndon Hill
Shropshire
WALES
England
NETHERLANDS
BELGIUM
GERM

INTRUSIVE LANDFORMS

6 BATHOLITH

Exposed Sierra Nevada Batholith

- It is a huge mass of igneous rocks, usually granite, which after removal of the overlying rocks form a massive and resistant upland region.
- Examples are Wicklow Mountains of Ireland, Uplands of Brittany, France and Main Range of West Malaysia.
- Their precise mode of origin is still a matter of controversy.
- It is generally believed that large masses of magma rising upwards metamorphosed the country rocks with which they came into contact.
- These metamorphosed rocks together with solidified magma gave rise to extensive batholiths.
- They are sometimes hundreds of kilometres in extent.
- They are more spectacular of the intrusive landform.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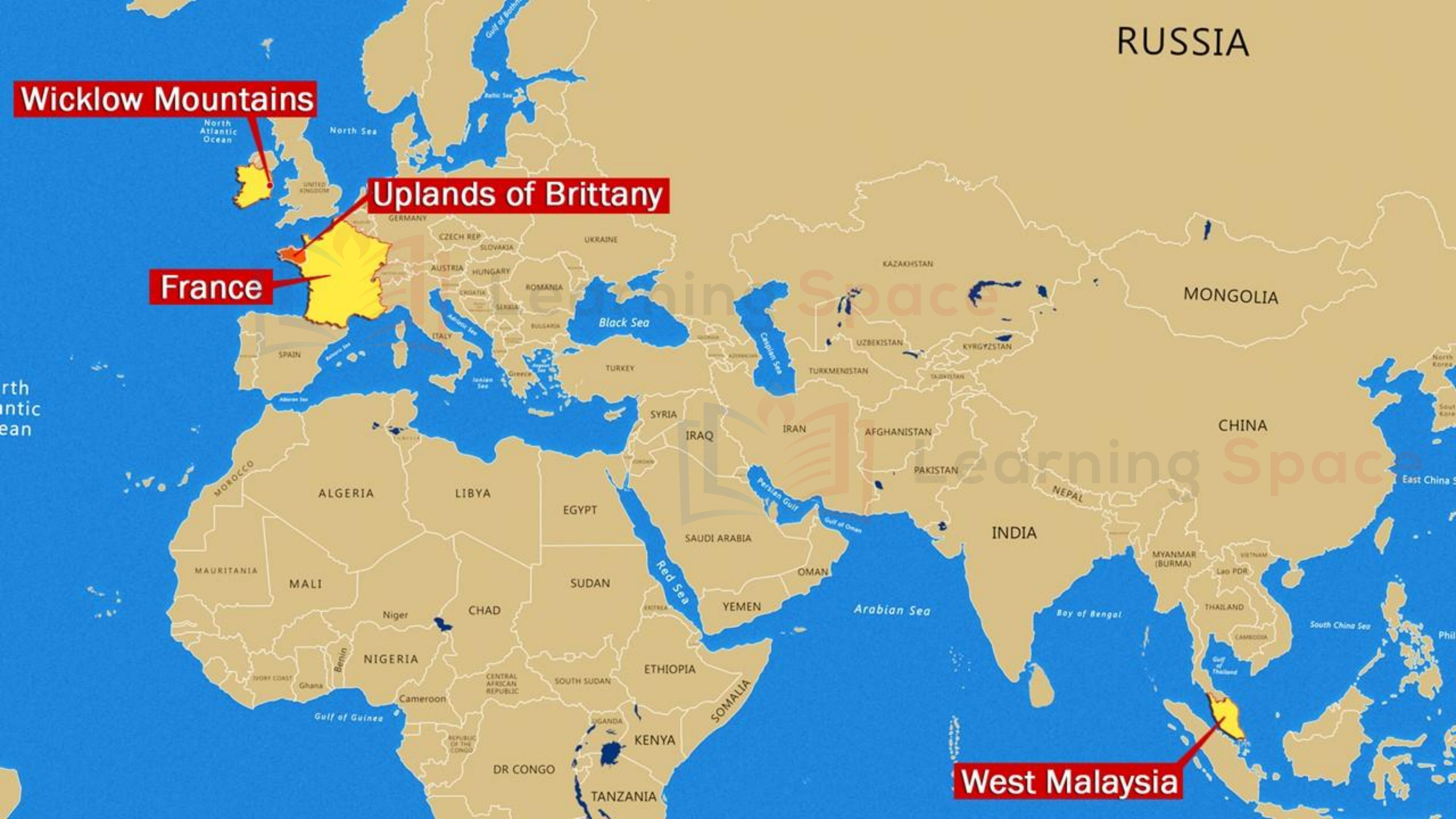
Wicklow Mountains
Uplands of Brittany
France
West Malaysia
RUSSIA
North Atlantic Ocean
North Sea
UNITED KINGDOM
GERMANY
CZECH REP
SLOVAKIA
UKRAINE
AUSTRIA
HUNGARY
ROMANIA
SERBIA
BULGARIA
ITALY
SPAIN
Greece
Black Sea
Caspian Sea
TURKEY
KAZAKHSTAN
UZBEKISTAN
KYRGYZSTAN
TURKMENISTAN
TAJIKISTAN
MONGOLIA
CHINA
SYRIA
IRAQ
IRAN
AFGHANISTAN
PAKISTAN
NEPAL
INDIA
MOROCCO
ALGERIA
LIBYA
EGYPT
SAUDI ARABIA
Persian Gulf
Gulf of Oman
OMAN
YEMEN
Red Sea
Arabian Sea
Bay of Bengal
MYANMAR (BURMA)
Lao PDR
VIETNAM
THAILAND
CAMBODIA
South China Sea
East China Sea
MAURITANIA
MALI
Niger
CHAD
SUDAN
NIGERIA
Benin
Ghana
IVORY COAST
Cameroon
CENTRAL AFRICAN REPUBLIC
SOUTH SUDAN
ETHIOPIA
SOMALIA
ERITREA
Gulf of Guinea
UGANDA
KENYA
REPUBLIC OF THE CONGO
DR CONGO
TANZANIA
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